

Grade 4 Math Lesson Plan

Chapter 2 – Lessons 2.1–2.2: Multiplication Comparisons & Comparison Problems

Duration: 45 minutes

Grade: 4

Subject: Mathematics

Topic: Multiplicative Comparison

Standards: 4.OA.A.1, 4.OA.A.2

1. Learning Objectives

By the end of the lesson, students will be able to:

- interpret expressions such as “**3 times as many as**” as multiplicative comparisons;
- represent multiplicative comparisons using a **bar model, comparison sentence, and equation**;
- identify the unknown quantity in a comparison problem;
- solve simple word problems involving multiplicative comparisons;
- distinguish between “**times as many**” and “**more than.**”

Success Criteria

Students can:

- [] explain what “times as many” means;
 - [] draw a comparison model correctly;
 - [] write a matching multiplication equation;
 - [] solve a comparison word problem and explain their reasoning.
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2. Key Vocabulary

- **factor**
- **product**
- **times as many**
- **comparison**
- **equation**
- **model**
- **difference**

Useful language frames:

___ is ___ times as many as ___.

___ $\times$ ___ = ___.

There are ___ times as many ___ as ___.

3. Materials

- Student Book, Chapter 2, Lessons 2.1–2.2
 - Whiteboard / projector
 - Mini whiteboards or paper
 - Markers / pencils
 - Bar-model worksheet or blank strips
 - Exit ticket
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4. Lesson Procedure

A. Warm-up – Activate Prior Knowledge

Time: 5 minutes

Display:

$$3 \times 5 = 15$$

Ask students:

1. What are the factors?
2. What is the product?
3. How could we describe this equation using words?

Introduce:

15 is 3 times as many as 5.

Then ask:

Is “3 times as many” the same as “3 more than”?

Use a quick example:

- 3 more than 5 = 8
- 3 times as many as 5 = 15

Emphasize that **multiplicative comparison is different from additive comparison.**

B. Mini-Lesson – Lesson 2.1: Multiplication Comparisons

Time: 10 minutes

Present the textbook-style problem:

Carly has 9 pennies. Jack has 4 times as many pennies as Carly. How many pennies does Jack have?

Step 1 – Identify the quantities

Carly = 9 pennies

Jack = 4 times Carly's amount

Step 2 – Draw a bar model

Carly:

[9]

Jack:

[9] [9] [9] [9]

Step 3 – Write an equation

$$n = 4 \times 9$$

$$n = 36$$

Therefore:

Jack has **36 pennies.**

Ask:

- What does the number **4** represent?
- What does the number **9** represent?
- What does **36** represent?

Key Concept

A multiplicative comparison tells us how many times greater one quantity is than another.

C. Guided Practice – From Model to Equation

Time: 8 minutes

Solve together:

There are 8 students in the art club. There are 3 times as many students in chorus. How many students are in chorus?

Students complete:

Model

Art club:

[8]

Chorus:

[8] [8] [8]

Equation

$$n = 3 \times 8$$

$$n = 24$$

Comparison sentence

There are 3 times as many students in chorus as in the art club.

Then reverse the representation.

Give students:

$$7 \times 5 = 35$$

Ask them to write a comparison sentence.

Expected:

35 is 7 times as many as 5.

or

7 times as many as 5 is 35.

D. Lesson 2.2 – Solving Comparison Problems

Time: 10 minutes

Move from direct comparisons to problems involving a **total**.

Present:

Maria's dog weighs 6 times as much as her rabbit. Together, the pets weigh 56 pounds. How much does Maria's dog weigh?

Explain that this problem is more challenging because the smaller amount is unknown.

Step 1 – Draw the model

Rabbit:

[n]

Dog:

[n] [n] [n] [n] [n] [n]

Total = 56

There are **7 equal parts** altogether.

Step 2 – Find one part

$$7 \times n = 56$$

$$n = 8$$

Rabbit = 8 pounds

Step 3 – Find the dog's weight

$$6 \times 8 = 48$$

Therefore:

Maria's dog weighs **48 pounds**.

Teacher Check

Ask:

Why did we divide the total into 7 equal parts instead of 6?

Expected understanding:

There are 6 parts for the dog plus 1 part for the rabbit.

E. Pair Challenge – Model, Solve, Explain

Time: 7 minutes

Students work in pairs.

Problem

At the zoo, there are 3 times as many monkeys as lions. There are 24 monkeys and lions altogether. How many monkeys are there?

Students must:

1. identify the smaller quantity;
2. draw a bar model;
3. write an equation;
4. solve;
5. explain their answer to their partner.

Expected solution

Lion:

[n]

Monkeys:

[n] [n] [n]

$$4 \times n = 24$$

$$n = 6$$

Monkeys:

$$3 \times 6 = 18$$

Answer: 18 monkeys

Extension

Ask early finishers:

How many more monkeys are there than lions?

$$18 - 6 = 12$$

F. Exit Ticket

Time: 5 minutes

Students complete independently.

Question 1

Write a comparison sentence for:

$$6 \times 4 = 24$$

Question 2

Ben has 4 times as many stickers as Anna. Anna has 5 stickers. How many stickers does Ben have?

Question 3

Tom has 3 times as many books as Mia. Together they have 20 books. How many books does Tom have?

Answers

1. 24 is 6 times as many as 4.
 2. $4 \times 5 = \mathbf{20 \text{ stickers}}$
 3. 4 equal parts = 20 $\rightarrow$ one part = 5 $\rightarrow$ Tom = $3 \times 5 = \mathbf{15 \text{ books}}$
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5. Assessment

Formative Assessment

During the lesson, observe whether students can:

- interpret “times as many” correctly;
- distinguish multiplicative from additive comparison;
- draw bar models;
- identify the unknown;
- translate models into equations.

Exit Ticket

Use the three exit-ticket questions to classify students:

Secure: 3/3 correct

Developing: 2/3 correct

Needs Support: 0–1/3 correct

6. Differentiation

Support

For students who need additional support:

- use concrete counters or cubes;
- provide partially completed bar models;
- highlight key phrases such as “**times as many**” and “**together**”;
- use smaller multiplication facts;
- provide sentence frames.

Example:

___ is ___ times as many as ___.

Challenge

For advanced students:

- provide problems where the smaller quantity is unknown;
 - include total-and-comparison problems;
 - ask students to calculate the difference between the quantities;
 - ask students to create their own comparison problem for a partner.
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7. Common Misconceptions

Misconception 1

Students interpret:

“4 times as many as 3”

as:

$4 + 3$.

Teacher response: Compare:

- 4 more than 3 = 7
 - 4 times as many as 3 = 12
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Misconception 2

Students multiply the total by the comparison factor in a Lesson 2.2 problem.

Teacher response: Return to the bar model and count the total number of equal parts first.

Misconception 3

Students confuse the larger and smaller quantities.

Teacher response: Ask:

“Which quantity has more equal parts?”

8. Homework / Optional GitHub Practice

Students can complete three additional problems:

1. 5 times as many as 6 is __.
2. There are 4 times as many red pencils as blue pencils. There are 3 blue pencils. How many red pencils are there?
3. There are 5 times as many cats as dogs. Together there are 24 animals. How many cats are there?

Answers

1. 30
 2. 12
 3. 20
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9. Teacher Reflection

After the lesson, consider:

- Could students distinguish **“times as many”** from **“more than”**?
- Did the bar models improve understanding?
- Which students could solve total comparison problems independently?
- Should Lesson 2.2 receive additional practice in the next lesson?